

1. A power company wants to predict the electricity demand in different regions based on past usage, temperature, and time of year. What type of machine learning approach would be suitable? Outline the steps to build the model

Model type: Regression

Step 1: Collect the csv file using `pd.read_csv`

Step 2: preprocess the data to remove null values & replace NA values

Step 3: Change the categorical values to numerical using one hot encoding

Step 4: Create independent & dependent values to predict the demand

Step 5: split the dataset into train & test sets respectively

Step 6: Built models using linear regression, or decision tree or random forest to predict the demand value

Step 7: Evaluate the model using R2 or RMSE methods

Step 8: Deploy the model using pickle file

2. An e-commerce platform wants to identify and filter out fake product reviews based on user behavior, review patterns, and sentiment analysis. What type of machine learning approach would be suitable? Outline the steps to build the model.

Model Type: Classification

Step 1: Collect csv file using `pd.read_csv`

Step 2: preprocess the data to remove null values & replace NA values

Step 3: Create independent & dependent variables to predict the fake reviews

Step 4: Split the dataset into train & test set respectively

Step 5: Build classification model using logistic regression, random forest, decision tree

Step 6: evaluate the model using F1, Recall, precision, Accuracy, confusion matrix

Step 7: Deploy the model using pickle file

3. A city traffic department wants to estimate traffic congestion levels based on GPS data, road construction reports, and weather conditions. What type of machine learning approach would be suitable? Outline the steps to build the model.

Model type: Regression

Step 1: Collect the dataset file using `pd.read_csv`

Step 2: preprocess the data to remove null values & replace NA values

Step 3: Change the categorical values to numerical using one hot encoding

Step 4: Create independent & dependent values to estimate the traffic congestion number

Step 5: split the dataset into train & test sets respectively

Step6: Built models using linear regression, or decision tree or random forest to estimate the traffic congestion

Step 7: Evaluate the model using R2 or RMSE methods

Step 8: Deploy the model using pickle file

4. A bank wants to determine whether a loan applicant should be approved or rejected based on income, credit history, and previous loan repayment behavior. What type of machine learning approach would be suitable? Outline the steps to build the model.

Model Type: Classification

Step 1: Collect dataset file using `pd.read_csv`

Step 2: preprocess the data to remove null values & replace NA values

Step 3: Create independent & dependent variables to predict whether application is selected or rejected

Step 4: Split the dataset into train & test set respectively

Step 5: Build classification model using logistic regression, random forest, decision tree to find out whether application is selected or rejected

Step 6: evaluate the model using F1, Recall, precision, Accuracy, confusion matrix

Step 7: Deploy the model using pickle file

5. A factory wants to automatically detect defective products using images from a quality control camera. What type of machine learning approach would be suitable? Outline the steps to build the model.

Model type: Classification

Step 1: Collect the data set of images

Step 2: Preprocess the images by image cleaning & normalization

Step 3: create independent & dependent variables to predict the defect images

Step 4: Split the data set into train & test set respectively

Step 5: Build classification model using logistic regression or random forest, or decision tree or svm or grid search or KNN

Step 6: evaluate the model using F1, Recall, precision, accuracy, confusion matrix

Step 7: Deploy the model using pickle file

6. An agricultural company wants to predict the best crops to grow based on soil composition, rainfall, and past harvest data. What type of machine learning approach would be suitable? Outline the steps to build the model.

Model type: Regression

- Step 1: Collect the agriculture dataset using `pd.read_csv`
- Step 2: preprocess the data to remove null values & replace NA values
- Step 3: Change the categorical values to numerical using one hot encoding
- Step 4: Create independent & dependent values to predict the best crops based on soil consumption
- Step 5: split the dataset into train & test sets respectively
- Step 6: Built models using linear regression, or decision tree or random forest to predict best crops
- Step 7: Evaluate the accuracy of model using R2 or RMSE methods
- Step 8: Deploy the model using pickle file

7. A business wants to analyze customer responses to different types of advertisements and promotions to determine the most effective marketing strategy. What type of machine learning approach would be suitable? Outline the steps to build the model.

Model Type: Classification

- Step 1: Collect advertisement dataset file using `pd.read_csv`
- Step 2: preprocess the data to remove null values & replace NA values
- Step 3: Create independent & dependent variables to predict which advertisements are effective
- Step 4: Split the dataset into train & test set respectively
- Step 5: Build classification model using logistic regression, random forest, decision tree to find out which advertisement determine effective marketing strategy
- Step 6: evaluate the model using F1, Recall, precision, Accuracy, confusion matrix
- Step 7: Deploy the model using pickle file

8. A software company wants to build a system that can analyze source code and predict whether a particular piece of code is likely to contain a bug. What type of machine learning approach would be suitable? Outline the steps to build the model.

Model Type: Classification

Step 1: Collect source code dataset file using `pd.read_csv`

Step 2: preprocess the data to remove null values & replace NA values, encode categorical values

Step 3: Create independent & dependent variables to predict which part of code contains a bug

Step 5: Build classification model using logistic regression, random forest, decision tree to find out which part of the code is a bug

Step 6: evaluate the model using F1, Recall, precision, Accuracy, confusion matrix

Step 7: Deploy the model using pickle file

9. A fitness app wants to recommend personalized workout plans for users based on their exercise history, fitness level, and preferences. What type of machine learning approach would be suitable? Outline the steps to build the model.

Model Type: Clustering

Step 1: Collect the fitness app dataset containing the user details, workout plans, using `pd.read_csv`

Step 2: preprocess the data to remove null values, encode categorical values into numerical values

Step 3: Build model using clustering methods like DBSCAN or Agglomerative clustering or hierarchical clustering to create the personalized work out plans

Step 4: Once the model is created, deploy it

10. A social media platform wants to detect fake accounts by analyzing user activity, posting patterns, and interactions with other users. Give steps to Achieve.

Model Type: Classification

Step 1: Collect social media dataset file containing user details, user activity, posts etc using `pd.read_csv`

Step 2: preprocess the data to remove null values & replace NA values, encode categorical values

Step 3: Create independent & dependent variables to predict whether user is fake or not

Step 5: Build classification model using logistic regression, random forest, decision tree to find out the fake account

Step 6: evaluate the model using F1, Recall, precision, Accuracy, confusion matrix

Step 7: Deploy the model using pickle file